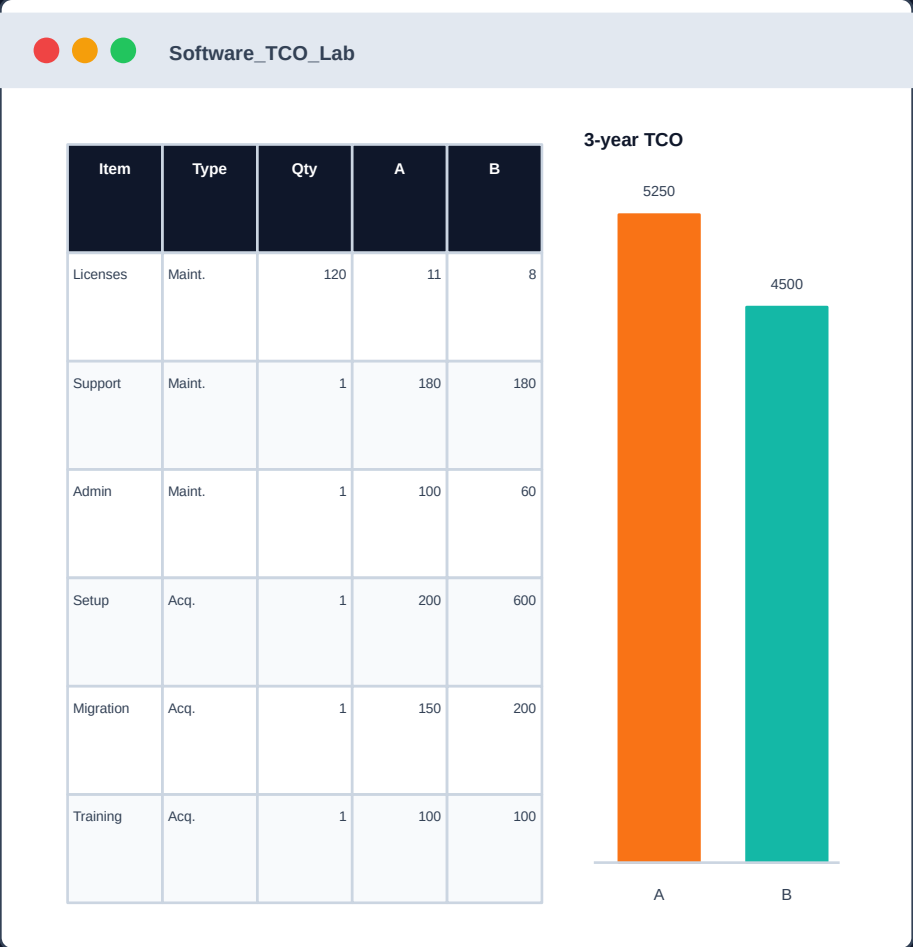


Utilizing Software to Estimate the Total Cost of Software Acquisition and Maintenance

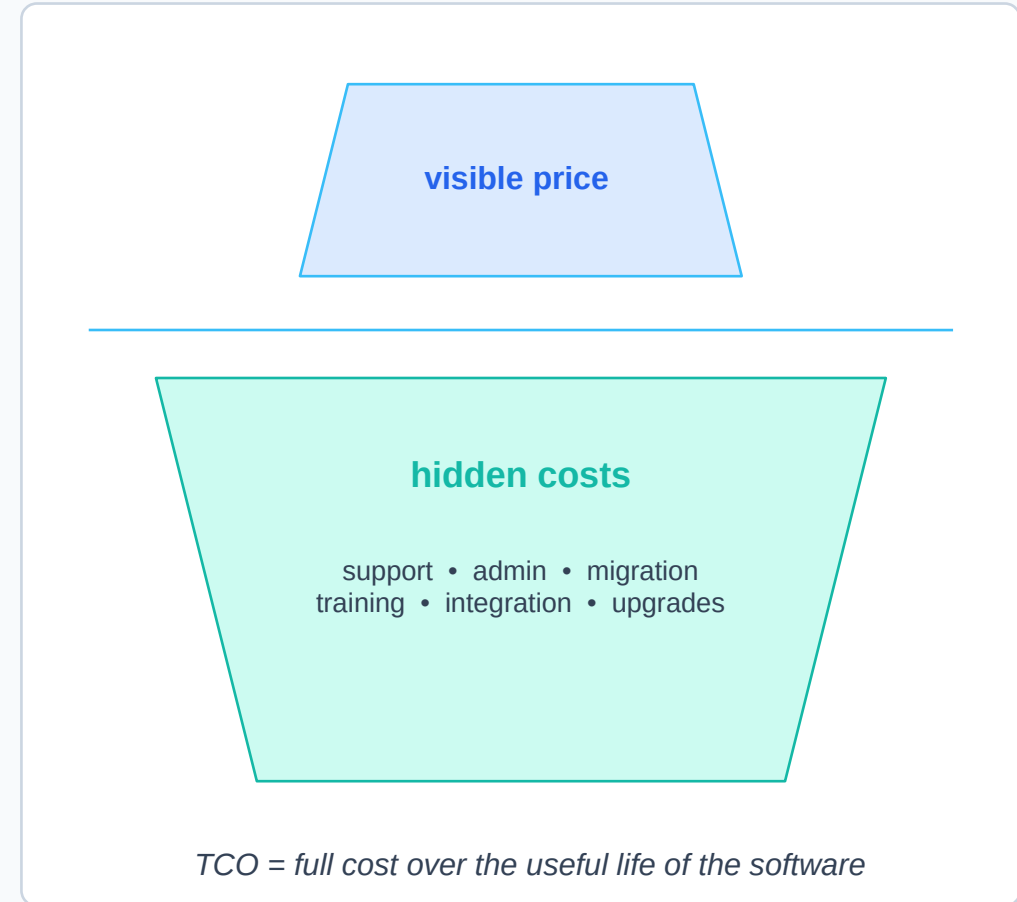
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26.05.2026



Why do we need software for TCO estimation?

Introduction

- Low purchase price does not always mean low lifecycle cost.
- Software decisions include both direct and indirect costs.
- Maintenance, support, migration and training are often underestimated.
- Project managers need comparable, repeatable and transparent estimates.
- A structured tool reduces manual calculation mistakes.



What costs should be included?

Scope of the estimate



Acquisition costs

- initial license or subscription commitment
- setup and configuration
- data migration/import
- initial training
- initial integration work



Maintenance costs

- recurring license/subscription renewals
- vendor support
- administration effort
- upgrades and changes
- refresher training or add-ons

$$\text{TCO} = \text{acquisition total} + \text{years} \times \text{annual maintenance total}$$

User needs — project manager perspective

Functional expectations

- List all one-time acquisition costs.
- List all recurring maintenance costs.
- Compare at least two software alternatives on the same basis.
- Update assumptions quickly and recalculate automatically.
- Present the result clearly to stakeholders.
- Keep track of assumptions and changes.



Functional Specifications

User needs translated into concrete software functionality



Functionality specification

MoSCoW prioritized requirements

Functionality	Input	Output	Purpose	MoSCoW
Cost item entry	item name, type, quantity, unit cost	structured cost record	capture all relevant cost components	Must
Formula calculation	numeric values + formulas	row/column totals	reduce manual errors	Must
Cost classification	Acquisition/Maintenance label	grouped cost categories	separate one-time from recurring costs	Must
Multi-year TCO	acquisition subtotal, annual maintenance, period	lifecycle total	support long-term decision-making	Must
Editable assumptions	changed users, prices, time	recalculated estimate	adjusting constants	Should
Summary view	subtotal and summary data	chart/dashboard view	communicate results to stakeholders	Should
Sharing/export	shared link, export	report-ready output	review and later report	Could

Market & Price Analysis

Comparable tools and pricing



Comparative price analysis

Programs analyzed

Program	Type	Producer	Price	Best fit / limitation
Google Sheets	Web	Google	Free personal; Business Starter \$7 per user/mo; Standard \$14 per user/mo (annual)	Fastest access; not a dedicated platform
Microsoft Excel	Desktop + Web	Microsoft	Business Basic \$6 per user/mo; Standard \$12.50 per user/mo (annual)	Strong spreadsheet depth; works best in Microsoft ecosystem
Smartsheet	Web	Smartsheet Inc.	Pro \$9 per member/mo; Business \$19 per member/mo (annual); Enterprise custom	Strong dashboards/reports; paid-plan dependency
Airtable	Web + desktop apps	Airtable	Free; Team \$20 per user/mo; Business \$45 per user/mo (annual)	Structured data apps

Which solution is the best fit for this class?

Suitability matrix

Criterion	Google Sheets	Excel	Smartsheet	Airtable
Cost item entry	Good	Good	Good	Good
Formula calculation	Good	Good	Medium	Medium
Cost classification	Good	Good	Good	Good
Multi-year TCO	Good	Good	Medium	Weak
Editable assumptions	Good	Good	Medium	Weak
Summary view	Good	Good	Good	Medium
Sharing/export	Good	Good	Good	Medium

Selected

**Google
Sheets**

Selected software — Google Sheets

- Starts immediately in the browser: type sheets.new.
- Free for personal Google accounts; Workspace business plans start from \$7 per user/month annual.
- Formulas support automatic TCO calculations and updates.
- Charts turn the estimate into a stakeholder-friendly result.
- Dropdowns, sharing and version history support controlled collaboration.



Practical Exercise

TCO comparison in Google Sheets



Software demonstration — setup

- 1 Open a browser.
- 2 Type sheets.new in the address bar.
- 3 Sign in with a Google account if needed.
- 4 Rename the file: Lab11.
- 5 Create one sheet named TCO.

Build the TCO table

Header row to create in Google Sheets

Cost item	Cost type	Quantity	Option A unit cost	Option A row total	Option B unit cost	Option B row total
Subscription licenses	Maintenance	120	11		8	
Vendor support	Maintenance	1	180		180	
Admin effort	Maintenance	1	100		60	
Setup/configuration	Acquisition	1	200		600	
Migration/import	Acquisition	1	150		200	
Initial training	Acquisition	1	100		100	

Dropdown setup: select B2:B7 → Insert → Dropdown → values: Acquisition, Maintenance

Calculate and visualize the result

Core formulas

E2 = C2*D2

G2 = C2*F2

B11 = SUMIF(\$B\$2:\$B\$7,"Acquisition",\$E\$2:\$E\$7)

B12 = SUMIF(\$B\$2:\$B\$7,"Maintenance",\$E\$2:\$E\$7)

B13 = B11+B12

B14 = B11+3*B12

Metric	Option A	Option B
Acquisition total	450	900
Annual maintenance	1600	1200
Year 1 total	2050	2100
3-year TCO	5250	4500

Tasks for the class

Task 1

Build a table in Google Sheets with 7 columns: Item, Type, Quantity, Unit cost A, Unit cost B, Total A, Total B. Add at least 5 entries to it. Make “Type” column a dropdown with 2 options: Acquisition and Maintenance. Write the correct formulas for each “Total A” and “Total B”

Task 2

Create a second small table with 3 columns: Type, Total A, Total B. Add 2 rows: “Acquisition” and “Maintenance”. Write the correct formulas using SUMIF.

Task 3

Create the final table with 3 columns: “Number of years”, “Total A” and “Total B”, and 3 rows: “1”, “5” and “10”. Write formulas that correctly calculate TCO for both options A and B in each time period.